

LEIGH ANNE MILLER

FORWARD DEPLOYED ENGINEER · AI SOLUTION ARCHITECT

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I'm a forward-deployed engineer: I embed with the customer, find the real problem, and **ship the AI system to production myself.**

\$40B/YR

AMPLIFY, HP's first API-based channel data exchange. Led enterprise data-strategy programs targeting \$15B in new revenue.

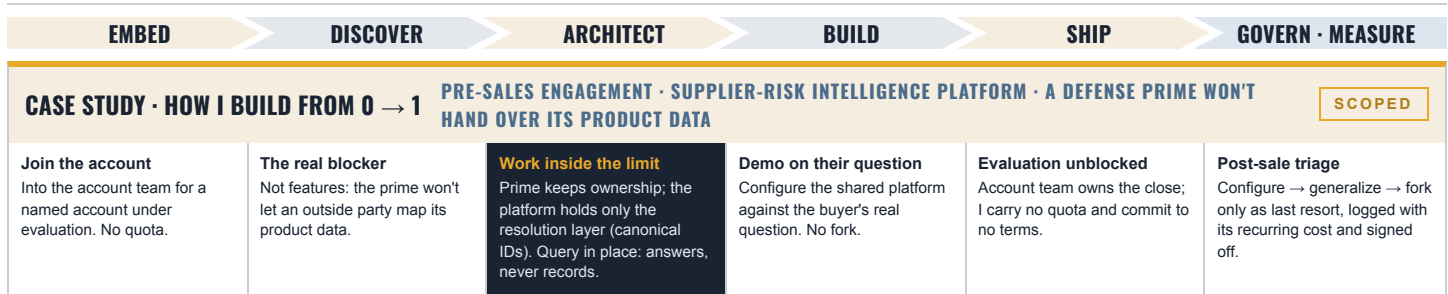
0 → 1

Multiple products opened from zero: discovery, demos, Tier-1 deployment, and the engineering.

\$300M

Unified data platform restarted and shipped at Nielsen/Gracenote after a five-year impasse. I supplied the revenue case.

THE FDE LOOP



FDE WORK: LIVE, IN PRODUCTION

BUILT SOLO · DIN · DAN IN PRODUCTION · EGON IN PILOTS

I OWNED, SOLO

Discovery

Architecture

Full stack

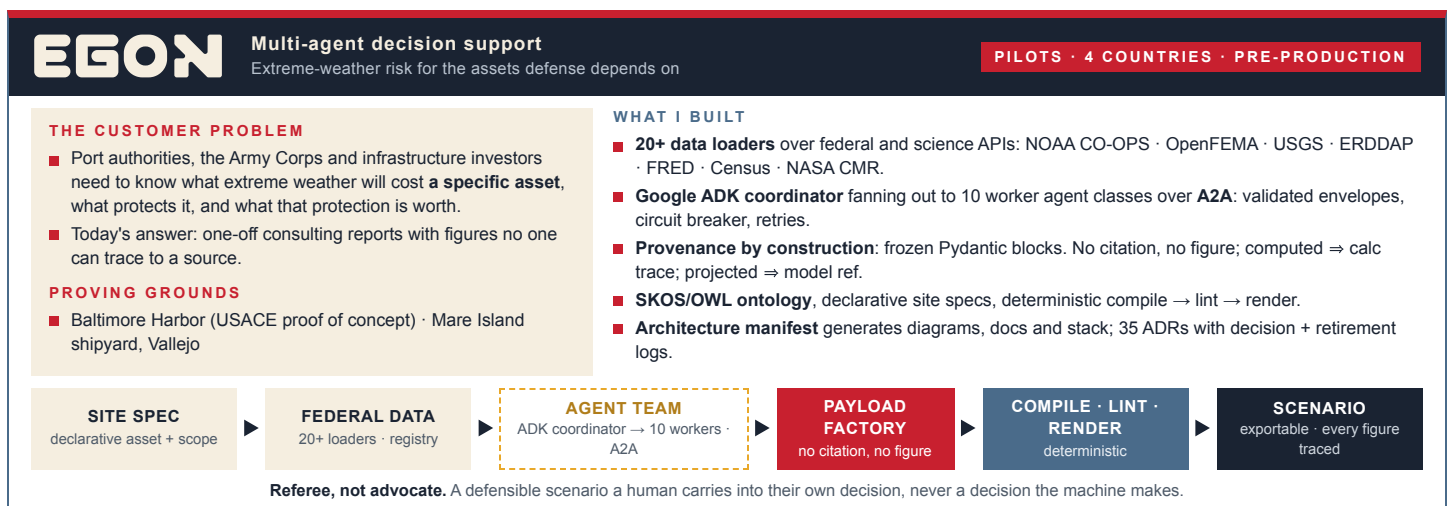
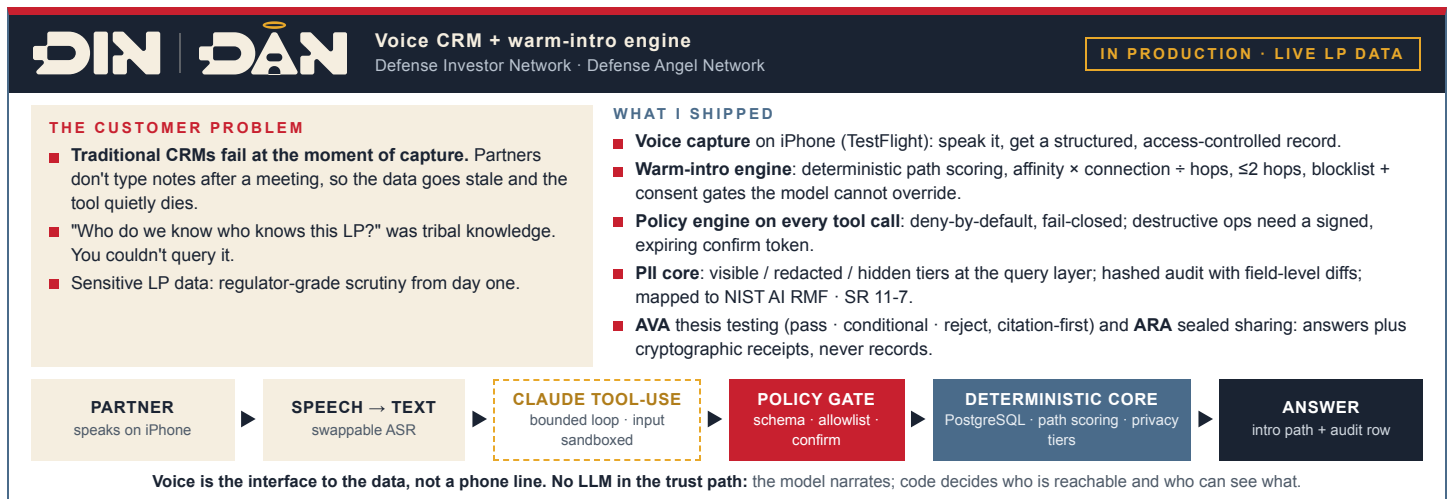
ML ops

Multi-tenant cloud

Security

Governance

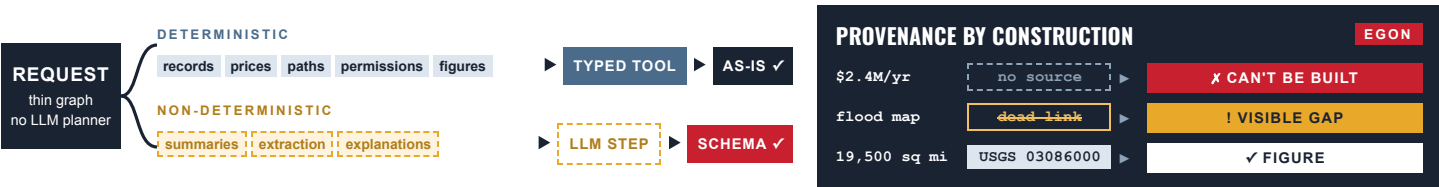
Rollout



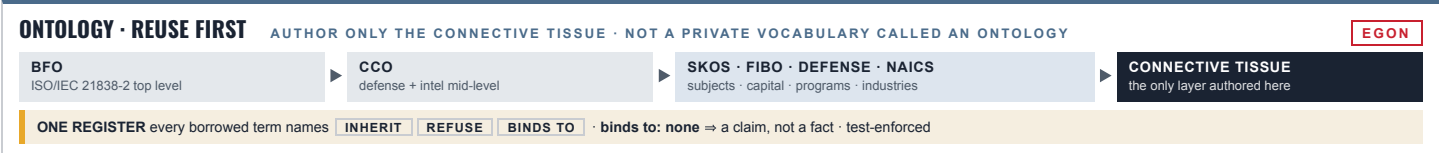
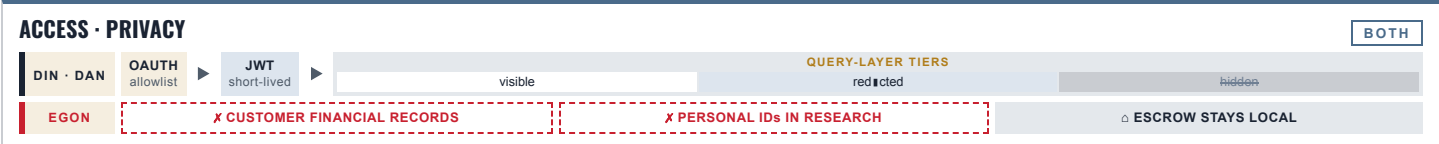
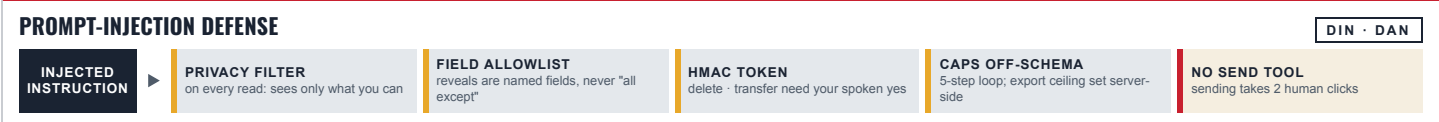
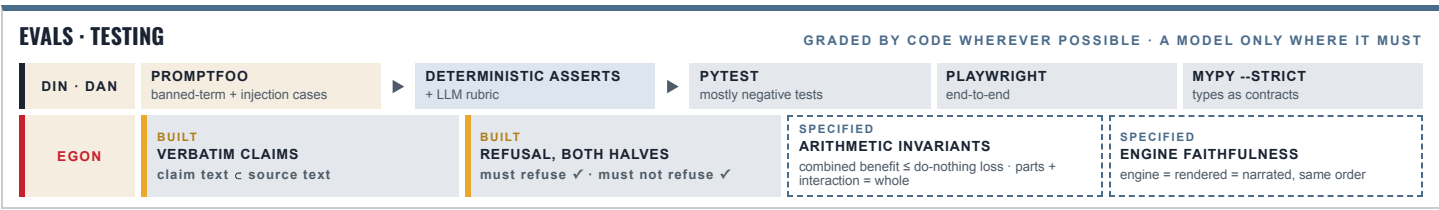
THE STACK WHAT I BUILD WITH

LANGUAGES	Python · TypeScript · Java · SQL
LLMS · AGENTS	Claude · Llama (local) · Claude Code · Google ADK · A2A · MCP · LangGraph
CONTRACTS · TESTS	Pydantic v2 (frozen models) · mypy --strict · pytest · Playwright · promptfoo
DATA	PostgreSQL/pgvector · SQLAlchemy · BigQuery · rdflib (SKOS/OWL) · Kafka
CLOUD · APP	GCP (Cloud Run · Cloud SQL · Vertex AI) · Confidential Space · AWS · FastAPI · React · Capacitor iOS
VOICE · TRACING	Google Chirp · ElevenLabs · Deepgram · Groq · Arize · OpenTelemetry

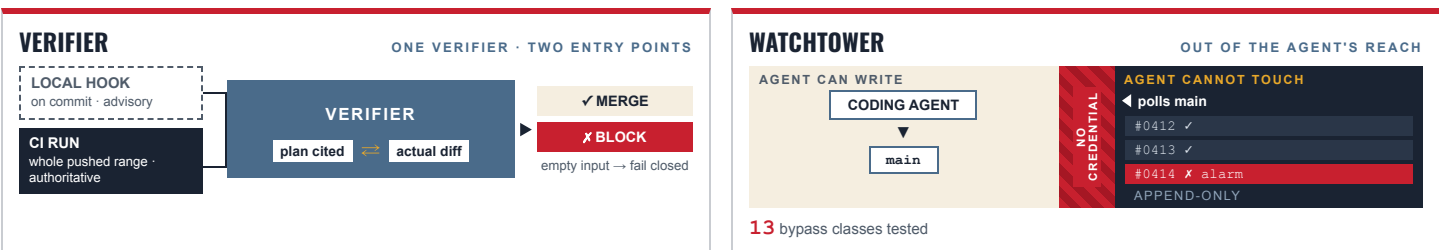
HOW I BUILD ENFORCED CONTRACTS · LIGHT DECLARATIVE ORCHESTRATION



PRODUCTION PATTERNS AND WHERE EACH ONE RUNS



WATCHTOWER BESPOKE TOOLING TO GOVERN AI AGENTS ON REGULATED, DATA-INTENSIVE BUILDS



FIELD RECORD CUSTOMER-FACING DEPLOYMENTS

2018–21	2021–23	2023–24	2024	2025–26	2026–	2026–
HP AMPLIFY: partner-facing channel data exchange, API-first \$40B/YR	Nielsen / Gracenote Unified data platform on AWS, uninstalled after a five-year impasse \$300M/YR	Ripley Decision Advantage Founder-engineer: compound AI systems, built and shipped solo	SpinF AI · BootstrapLabs Real-time voice AI on live sales calls, discovery through Tier-1 deployment 0 → 1	X, The Moonshot Factory AI engineer	DIN · DAN AI Solution Architect. Voice CRM + intro engine on live LP data LIVE	Ripley · EGON Multi-agent decision support for ports and federal agencies PILOTS

Q WHAT'S UNDER THE VOICE INTERFACE?

DIN · DAN



Q AN EXTERNAL SERVICE IS SLOW OR FAILING?

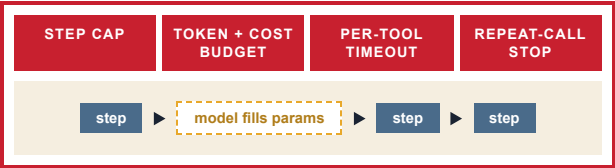
Never a 30-second spinner. Never stale data shown as fresh.



No retry on a bad 400 · idempotency keys on writes · long jobs go async: job ID, then poll or push.

Q HOW DO YOU STOP RUNAWAY TOOL LOOPS?

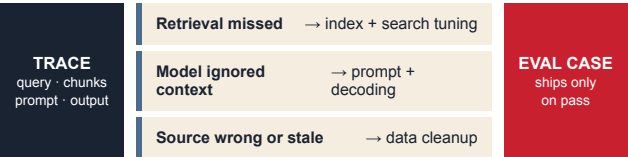
The graph decides what runs next. The model only fills in parameters.



A limit hit stops the run and escalates to a human, with the full trace.

Q THE SYSTEM GAVE A WRONG ANSWER. NOW WHAT?

Classify the failure first. Each cause has a different fix.



Q AUTHENTICATION AND AUTHORIZATION?

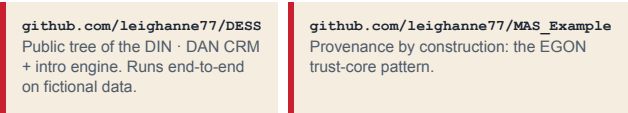
Identity is short-lived; access is checked on every request and logged.



Services: workload identity, mTLS, rotated secrets. DIN · DAN Google OAuth allowlist -> JWT; privacy tiers at the query layer.

Q HOW HANDS-ON ARE YOU?

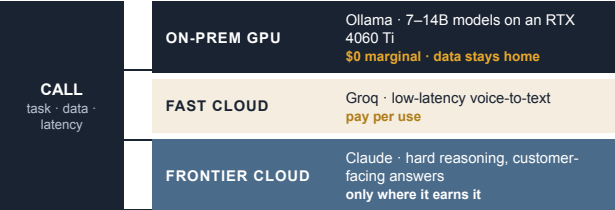
I code daily, Python first. Don't take my word for it:



```
$ git clone ... && pytest · FastAPI · LangGraph · ADK · Node · Java
```

Q HOW DO YOU MANAGE MODEL COSTS?

Route each call to the cheapest place that meets the bar, and meter everything.



On-prem runs coding assist, the local Llama judge and bulk jobs. DIN · DAN per-user token + voice budgets; web-search spend capped and audited. Right-size infra too: PostgreSQL, not Spark.